

MIGUEL MACEDO

AI DATA ENGINEER

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PROFILE

Data Engineer with 10+ years at Celfocus, with recent focus on GCP/BigQuery migration, data governance and GenAI-enabled workflows.

KEY SKILLS

- Python
- SQL & PL/SQL
- Google Cloud Platform (BigQuery)
- n8n (Workflow Automation)
- Teradata & BTEQ
- Oracle BRM
- Oracle ODI (ETL)
- Oracle Forms & Reports
- Shell Scripting
- Talend Open Studio
- Git & Jenkins
- Agile / SAFe

CORE COMPETENCIES

- Software Design & Analysis
- Database Design & Development
- Billing Systems
- Customer Relationship Management
- Technical Troubleshooting
- Data Governance & Compliance

LANGUAGES

Portuguese — Native
English — Advanced (Written & Verbal)
Spanish — Basic

EDUCATION

Bachelor's Degree, Information Systems & Technology
Universidade do Minho
Sep 2006 – Jul 2011

TRAINING & CERTIFICATIONS

Automation & Agents with Artificial Intelligence
codeForAll — 2026
Hands-on RAG pipelines & n8n workflows

Oracle Data Integrator (ODI 12c)
Udemy — 2024

PSM I
Scrum.org — 2023

ITIL v4
Actual Training — 2020

INTERESTS

Music — performing in a variety of settings and environments
Tennis (federated, 16 years) and Padel — competitive team & individual sport

PROFESSIONAL SUMMARY

Data Engineer with 10+ years of experience at Celfocus, spanning analyst, developer and support roles across billing, ETL and cloud migration projects. Skilled in building BI and data warehousing solutions and developing Python- and PL/SQL-based tooling. Currently supporting Vodafone Germany's migration from Teradata to Google BigQuery, with hands-on training in RAG-based AI workflows using n8n.

PROFESSIONAL EXPERIENCE

Data Engineer — Celfocus — Vodafone Germany (BIT Project)

Oct 2025 – Present

- Delivered within a SAFe Agile framework, contributing to the Simplification and Semantic teams' planning and delivery cadence.
- Simplification team: replicated legacy data warehouse tables into the new Teradata data warehouse, unit-tested each migration, wrote BTEQ scripts and produced evidence of data integrity to support deployment.
- Semantic team: developed, tested and maintained solutions supporting the migration from legacy Teradata to Google BigQuery.
- Built a Python command-line module to create a table or a list of tables directly in BigQuery, or generate their DDL.
- Developed a Python solution aligning Semantic tables with Vodafone's internal permissioning process — extracting table metadata from JSON files and generating the corresponding SQL INSERT statements.

Technologies: Python, Google Cloud Platform (BigQuery), Teradata, BTEQ, Git, VSCode

Analyst / Developer / Support — Celfocus — Billing & IITC

Jul 2017 – Oct 2025

- Billing Dev Team: helped evolve the design of the EBUA solution, performed testing, and acted as SPOC between the Development and QA teams; supported the customer after go-live.
- IITC team: developed BI reports and ODI (ETL) processes, and served on the 3rd-line support team.
- Designed and built a new ETL pipeline for CDRs sourced from MediationZone/Openet (ECS), developing new ODI processes and PL/SQL procedures, functions and packages.
- Participated in the 24/7 on-call support rotation throughout the engagement.

Technologies: Oracle ODI, Oracle Database, Oracle PL/SQL, Oracle BRM, Oracle BI Publisher, Shell Scripting

Oracle Developer — Celfocus — Vodafone Turkey (CBU Migration)

Jul 2016 – Jul 2017

- Supported the team in building the CBU migration solution using SQL, PL/SQL and Talend Open Studio, including identifying duplicate index entries across phased deployments.

Technologies: Oracle Database, Oracle PL/SQL, Talend Open Studio

Oracle Developer — SPMS — Health Ministry

Jul 2014 – Jul 2016

- Developed and maintained the national clinical software used by physicians, evolving it from SAM to SClinico and upgrading from Oracle Forms 6i to 10g; implemented the first version of the software used by Nutritionists.

Technologies: Oracle Forms & Reports (6i & 10g), Oracle Database, Oracle PL/SQL

Global Support Analyst (1st-3rd Line) — SPMS — Health Ministry

Sep 2012 – Jul 2014

- Helped establish the Support Centre for all Healthcare solutions at the Health Ministry, creating procedures to help physicians resolve software issues.

Oracle Developer — Glintt HS

Jan 2012 – Sep 2012

- Developed and maintained Glintt's healthcare software solution, built on Oracle Forms & Reports, Oracle Database and PL/SQL.